

Sujay Seeram

Software Systems, AI Infrastructure & High-Performance Runtimes Engineer

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ENGINEERING PROFILE & CORE PHILOSOPHY

Undergraduate systems and AI engineer at Amrita Vishwa Vidyapeetham with an established track record in **high-performance distributed runtimes**, **sovereign AI architectures**, and **cyber-physical systems**. An open, execution-oriented learner and quick adapter driven by the “**Extreme Economic Engineering**” paradigm: deriving minimalist algorithmic and compute configurations from mathematical first principles to drastically slash latency, memory footprint, and cloud dependency. Beyond core engineering, deeply committed to applying technological innovation toward **renewable energy**, **environmental sustainability**, and **social responsibility**. Disciplined Naval NCC cadet and 3-time Services Selection Board (SSB) attendee.

EDUCATION

- **Amrita Vishwa Vidyapeetham** Coimbatore, India
Bachelor of Technology in Electronics and Communication Engineering (ECE) *Expected 2027*
 - **Core Focus:** High-Throughput I/O Runtimes, Local Vector Indexing, Physics-Informed ML, Autonomous Agent Protocols.
- **Navy Children School (NCS), CBSE** Visakhapatnam, India
Senior Secondary (Class XII): 92.5% | Secondary School (Class X): 95.0% *2021 – 2023*

TECHNICAL SKILLS

- **Programming Languages:** Python, Rust, TypeScript, JavaScript, C/C++17, Go, Dart (Flutter), SQL, POSIX Shell/Bash.
- **Systems, Networking & Runtimes:** Actix-Web, Tokio async, FastAPI, Node.js, WebSockets, PTY orchestration, Linux eBPF, Docker, Linux IPC.
- **Machine Learning, DSP & Retrieval:** PyTorch, Scikit-Learn, LanceDB (Vector Indexing), SQLite Local-RAG, Model Context Protocol (MCP), Gemini API.
- **Frontend & UI Systems:** React 18, Vite, React Native, Three.js / React Three Fiber, WebGL, Reanimated, Streamlit.
- **Cloud, Security & Databases:** Google Cloud Run, Firebase Auth (OIDC), Cloud Firestore, PostgreSQL, Redis, Android Accessibility API Security.

ACHIEVEMENTS, DISTINCTIONS & MILITARY DISCIPLINE

- **Rapid Prototyping Portfolio & Engineering Track Record** *Portfolio Metrics*
 - **95 Projects Explored, 50 Executed, 3 Major Hackathons Won:** Interdisciplinary distribution of **15 AI+CS+ECE** cyber-physical projects, **25** dedicated ECE/embedded systems, **5** core CS/systems software, and **24** foundational sandbox builds.
 - **Real-Time Production Deployment (SAFE):** Engineered **SAFE**, an operational, production-grade warehouse asset tracking ecosystem featuring low-power BLE duty cycling and high-speed batch scanning (*details reserved for interview discussion*).
- **IIT Madras Smart India Challenge – 2nd Runner Up & Youngest Panelist** *IIT Madras*
 - Awarded **2nd Runner Up** at IIT Madras Smart India Challenge for developing high-efficiency distributed and cyber-physical architectures.
 - Invited and served as the **Youngest Panelist** at the summit, presenting on low-overhead systems and sovereign AI computation.
- **National Cadet Corps (NCC) & Services Selection Board (SSB)** *Armed Forces Selection*
 - **NCC Senior Division:** Trained cadet in **3(A) Naval Unit NCC, Visakhapatnam (VSP)**; developed crisis leadership, tactical problem-solving, and disciplined execution.
 - **Services Selection Board (SSB):** Attended 3 SSB selection boards (**Screened-in twice & Conferenced-out twice**; Screened-out once); demonstrated high situational resilience and decision-making under stress.
- **Global Automotive Research Centre (GARC) Technical Delegation** *Chennai, India*
 - Selected technical delegate at the **GARC Chennai Meet**; engaged on real-time telemetry pipelines, vehicle sensor failure preemption, and edge computing diagnostics.

FEATURED ENGINEERING PROJECTS (ORDERED PORTFOLIO)

I. Interdisciplinary Cyber-Physical AI (AI, CS & EC Combined)

- **1. EVolvAI – Generative AI & Physics-Constrained EV Grid Optimization** *Release v1.0.0*
 - Architected a scalable cyber-physical demand framework in PyTorch coupling a **Generative Counterfactual VAE (GCD-VAE)** with a Multi-Objective Genetic Algorithm to simulate charging loads under distribution grid bus limits and climate stress anomalies.
 - Achieved $R^2 \approx 0.5$ predictive accuracy on sparse field data; built an interactive Streamlit simulation console for municipal engineers.
- **2. MotorSafe – Tri-Modal Edge-AI Predictive Maintenance Ecosystem** *Release v0.01*
 - Constructed an edge-native telemetry pipeline fusing 3 synchronized sensor channels: **ACS712** current, **ZMPT101B** voltage, and **MPU6050** vibration.
 - Engineered on-chip FFT DSP to extract spectral sidebands and deployed an unsupervised **Isolation Forest** on ESP32 to preempt bearing breakdown without cloud latency, streaming live diagnostics to an interactive Three.js 3D digital twin dashboard.
- **3. SpecRAG – Zero-Hallucination Firmware & Datasheet Auditing Gateway** *Release v0.1.0*
 - Engineered an enterprise developer tool bridging 1,000-page silicon datasheets with embedded C/C++ firmware using the **Model Context Protocol (MCP)**.
 - Performed localized vector retrieval against indexed silicon register banks to preemptively catch peripheral pin collisions, clock-tree prescaler errors, and timing hazards directly in VS Code before compilation.

II. Embedded Systems & Electronics (ECE)

- **4. AQUAPULSE – Cognitive Software-Defined Sonar Payload for AUVs** *Repository*
 - Engineered a cognitive software-defined sonar payload for Autonomous Underwater Vehicles (AUVs) meeting the SIH26058 MoES mandate.
 - Implemented zero-CPU DMA-to-DAC acoustic synthesis on **STM32 / ESP32-S3** with **OPA1612** active filtering, deploying an INT8 **TinyML** engine to dynamically adapt ping frequency (100–480 kHz) across shifting thermoclines, slashing acoustic power consumption by **38%**.
- **5. Pheno-Speak – Edge Acoustic Vocal Biomarker Extraction Engine** *Repository*
 - Built an edge audio digital signal processing pipeline for non-invasive vocal biomarker extraction aligned with SDG 3.4 (Digital Phenotyping).
 - Extracted MFCCs, jitter, and shimmer features on lightweight microcontroller hardware without offloading unencrypted audio to external endpoints.
- **6. Dust-Vision & Dust-Sim – Edge Particulate Tracking & Dispersion Engine** *Release v0.1*
 - Deployed computer vision models on edge camera modules to quantify real-time construction site particulate pollution and dust plumes.
 - Coupled sensor data with a custom particle kinematics physics simulation (**dust-sim v1.0.0**) modeling dispersion under thermal and wind shifts.

III. Systems Software, Developer Tools & Computer Science (CS)

- **7. ytstr – High-Performance Psychoacoustic Audio Engine** *Release v2.0.0*
 - Developed a terminal-native streaming runtime powered by a psychoacoustic **DecisionEngine** executing real-time spectral analysis.
 - Implemented 8 broadcast-grade transition algorithms and built an active backpressure memory pool, slashing runtime RAM by **70%** (< 260MB).
- **8. Informed-Poll – Swipeable Civic Intelligence Platform** *Release v1.0.0*
 - Developed a mobile-first civic intelligence companion featured in the Google Solution Challenge, pairing swipe mechanics with verified candidate records.
 - Integrated localized neural RAG using **LanceDB** and Gemini 3.1 Flash Lite over a Cloud Run FastAPI backend with zero-hallucination verification.
- **9. Kavach-UPI – Android Accessibility Security Shield** *Repository*
 - Authored an on-device security shield leveraging Android Accessibility APIs to detect active screen-sharing malware during financial UPI transactions.
 - Instantly deployed an opaque system window overlay to block remote touch injections while triggering localized security alerts.